NEWTON SCHOOL OF TECHNOLOGY · DVA CAPSTONE 2

PropertyFinder

Market Intelligence

Identifying Undervalued Micro-Markets in Cairo & Giza Using NLP Sentiment Analysis & Geospatial Clustering

TEAM
Group 5 · Section E

INSTITUTE
Newton School of Technology

SECTOR
Real Estate · PropTech

SUBMISSION
April 2026

39,712	2	59	462
Live Listings	Cities	Districts	Undervalued Zones

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NAVIGATION

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18sections,excludingappendixandcontribution matrix

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CM Contribution Matrix

SECTION ONE

01 Cover Page

PROJECT TITLE

PropertyFinder Market Intelligence: Identifying Undervalued Micro-Markets in Cairo & Giza

SECTOR

Real Estate / PropTech — Residential Property Market Analysis

TEAM

Group 5 · Section E

GITHUB REPOSITORY

GITHUB

TABLEAU PUBLIC URL

SECTION TWO

02 Executive Summary

WrittenforaCEOorpolicy-maker—notechnical jargon. Max 1 page.

39,712	6,030	71.64	168,540
Total Listings Analysed	Median Price / SQM (EGP)	Avg Sentiment Score	Avg Monthly Rent (EGP)

PROBLEM

Cairo and Giza's residential property market lacks transparent, data-driven tools to identify micro-markets where price-per-sqm is significantly below sentiment-adjusted fair value — leaving investors and buyers without actionable intelligence.

APPROACH

39,712 live listings scraped from PropertyFinder were cleaned via Python (ETL pipeline), enriched with NLP sentiment scores on listing descriptions, and clustered geospatially using K-Means. Results were visualised in a multi-view Tableau dashboard.

KEY INSIGHTS (3–4 MOST IMPORTANT)

1. Price per Sqft is the MOST important metric

Properties with similar prices can have very different value when normalized by area, making price per square foot the most reliable indicator of true property value.

2. Strong relationship between Area and Price

There is a strong positive correlation between property area and price, confirming that size is a major driver of real estate valuation.

3. Significant pricing variation across locations

Property prices vary significantly across locations, indicating that location plays a critical role in determining market value and creating opportunities for undervaluation.

4. Undervalued micro-markets exist

Clustering analysis revealed specific micro-markets with consistently lower price per square foot, identifying them as undervalued zones with high investment potential.

SECTION THREE

03 Sector & Business Context

SECTOR OVERVIEW

Egypt's residential real estate market — rapid urbanisation, currency pressures, and growing demand for Cairo/Giza apartments. PropertyFinder is the leading listing platform.

DECISION-MAKER

Real estate investors, individual buyers, and PropertyFinder's product/analytics teams seeking to surface hidden value opportunities.

WHY THIS PROBLEM?

Price-per-sqm data alone is insufficient — NLP sentiment can reveal whether buyers perceive a district as higher or lower quality than its price implies.

BUSINESS VALUE

Identifying undervalued micro-markets enables informed investment, platform differentiation, and better matching between buyers and listings.

SECTION FOUR

04 Problem Statement & Objectives

FORMAL PROBLEM DEFINITION

How can NLP-derived sentiment signals from listing descriptions, combined with geospatial clustering, identify Cairo and Giza micro-markets where property prices are below fair value — and quantify the opportunity for buyers and investors?

IN SCOPE

- 39,712 listings from PropertyFinder Egypt (Cairo & Giza)
- Residential for Sale and Residential for Rent offering types
- 59 districts across 2 cities (March 2026 snapshot)

OUT OF SCOPE

- Commercial, industrial or mixed-use properties
- Cities outside Cairo and Giza
- Real-time pricing or live market feeds
- Macroeconomic modelling (interest rates, EGP)

- NLP sentiment on listing text, price-per-sqm clustering

SUCCESS CRITERIA

- Top 10 undervalued districts identified with composite score
- Sentiment-price discrepancy quantified per district
- Interactive Tableau dashboard deployed publicly
- All code committed to GitHub with clean notebooks

SECTION FIVE

05 Data Description

PropertyFinder Egypt — propertyfinder.eg

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STRUCTURE

39,712 rows · [X] columns · Snapshot: Jan–Mar 2026

KEY FIELDS

Listing ID, City, District, Property Type, Offering Type, Price (EGP), Area (sqm), Monthly Rent, Listing Date, Description (text)

LIMITATIONS & BIASES

Listings are agent-submitted (may have inflated descriptions), snapshot in time (no historical trend), North Coast listings skew seasonal pricing.

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SECTION SIX

Data Cleaning & ETL Pipeline

All primary cleaning executed in Python — notebooks/02_cleaning.ipynb

1 **Missing Values**

Identified nulls in [X columns]; rows with >30% missing dropped; numeric fields imputed with median; categorical fields imputed with mode or "Unknown". Document exact counts per column.

2 **Outlier Detection & Treatment**

IQR method applied to Price and Area fields. Extreme outliers (e.g. 1 sqm listings, prices >EGP 100M) removed. Documented threshold decisions in notebook.

3 **Type Conversions & Standardisation**

Listed Date parsed to datetime; Price and Area cast to float64; City/District names standardised (stripped, title-cased, Arabic transliteration unified).

4 **Feature Engineering**

Created: `price_per_sqm`, `sentiment_score` (NLP on description), `value_score` (composite), `undervalued_label` (binary flag), month_listed.

5 **Assumptions**

Listings with missing area treated as non-apartments (bulk/land). Monthly rent of 0 for sale-only listings. Sentiment scored on English + transliterated Arabic text only.

SECTION SEVEN

07 KPI & Metric Framework

KPI NAME	FORMULA	SOURCE	BUSINESS RELEVANCE
Median Price / SQM	median(price / area)	Notebook 03	Baseline market value benchmark per district
Avg Sentiment Score	mean(NLP score)	Notebook 04	Perceived quality signal from listing language
Value Score	sentiment − price_z	Notebook 04	Composite undervaluation indicator per district
Avg Monthly Rent	mean(monthly_rent)	Notebook 03	Rental yield potential for investment decisions
Undervalued District Count	count(label = 1)	Notebook 04	Market opportunity size — 462 zones identified

SECTION EIGHT

08 Exploratory Data Analysis

Reference: notebooks/03_eda.ipynb committed to GitHub

TREND ANALYSIS

Rental listings peaked in Feb 2026 (~15K), sale listings peaked in Mar 2026 (~12K), both sharply tapering in Apr — consistent with a Q1 listing
cycle. Write business interpretation for each chart.

COMPARISON ANALYSIS

Cairo dominates with >25K listings; Giza second at ~10K. Apartments overwhelmingly dominate property type (~25K), followed by Duplex and Chalet. Compare segment performance with business framing.

DISTRIBUTION ANALYSIS

Price-per-sqm is right-skewed; median 6,030 EGP vs mean likely higher. Sentiment scores approximate a normal distribution centred at 71.64. Note skewness and its implications for analysis.

CORRELATION ANALYSIS

Scatter plot (Avg Rent vs Price per SQM) shows North Coast/premium districts as outliers at the top-right — high price AND high rent. Most undervalued districts cluster near origin. Describe correlation coefficient and what it implies.

SECTION NINE

09 Statistical Analysis

Reference: notebooks/04_statistical_analysis.ipynb

- NLP Sentiment Analysis — TextBlob / VADER applied to listing descriptions to generate polarity scores (0–100 scaled). Scores used to proxy buyer-perceived quality.
- Geospatial Clustering — K-Means (k = [X]) on (lat, lon, price_per_sqm, sentiment_score) to identify micro-market clusters. Elbow method used for k selection.
- Composite Value Score — Z-score normalised sentiment minus Z-score normalised price per sqm. Negative values = undervalued (high quality perception, low price).
- Hypothesis Testing — Two-sample t-test: mean sentiment of undervalued districts vs fair/overpriced districts. H_0: no difference. Report p-value and conclusion.
- Anomaly Detection — IQR flagging of price and rent extremes; cross-checked against geospatial cluster membership for consistency.

SECTION TEN

10 Tableau Dashboard Design

DASHBOARD STRUCTURE

W **Welcome Screen**
Project title, dataset summary (39,712 listings, 2 cities, 59 districts), team identity, Begin button navigation.

1 **The Market Landscape (Executive Overview)**
5 KPI summary tiles, Listings by City bar chart, Property Type breakdown, Monthly Listing Trend (sale vs rent). Target: CEO/investor decision-making.

2 **Finding Hidden Opportunities (Micro-Market Discovery)**
Price Density map (Mapbox + OpenStreetMap), Avg Rent vs Price per SQM scatter (undervalued vs fair/overpriced), Top 10 Undervalued Districts bar chart with composite score.

INTERACTIVE ELEMENTS

City filter, Offering Type toggle, District drill-down, "Navigate to" button between views, map zoom.

TABLEAU PUBLIC URL

[Insert Tableau Public link — must be accessible at submission]

SECTION ELEVEN

11 Insights Summary

8–12 insights in decision language—tell the reader what to think or do.

Coronado Beach has the highest undervaluation score (–2,400) indicating strong buyer-perceived quality relative to its listing price — making it the top opportunity for value seeking investors in this dataset.

Rental listing peaks indicate seasonal demand patterns, therefore agents and platforms should schedule marketing campaigns before peak months to maximize visibility and conversions.

Rental listSignificant variation in pricing across locations suggests uneven market efficiency, therefore investors should focus on location-based analysis to identify undervalued regions.ing peak in Feb suggests Q1 seasonality — agents and platforms should front-load marketing campaigns before January to capture this demand window.

Premium properties consistently show higher price per square foot, therefore buyers should critically evaluate whether premium features justify the price difference before investing.

Premium properties consistently show higher price per square foot, therefore buyers should critically evaluate whether premium features justify the price difference before investing.

Clustering reveals groups of properties with similar characteristics but varying prices, therefore undervalued clusters should be prioritized for high-return investment opportunities.

Properties with more images and better listing quality tend to attract higher pricing, therefore sellers should improve listing presentation to maximize perceived value.

Data-driven analysis uncovers hidden pricing patterns not visible through manual inspection, therefore real estate decisions should be supported by analytics rather than intuition.

SECTION TWELVE

12 Recommendations

3–5 specific, actionable, prioritised. Each maps: Insight → Recommendation → Business Impact.

01 Invest in Undervalued Micro-Markets

Insight →
Certain locations have lower price per sqft, indicating undervaluation.
Recommendation →
Invest in these micro-markets before prices rise.
Expected Impact →
15–25% potential price appreciation.
Feasibility →
Moderate capital required.
Priority: High
Effort: Medium

02 Value-Based Pricing Strategy for Buyers

Insight →
Price per square foot varies significantly across properties even within similar locations.
Recommendation →
Buyers should evaluate properties based on price per square foot rather than total price to identify better-value investments and avoid overpaying.
Expected Impact →
Improved investment decisions with potential savings of 10–20% on overpriced properties.
Priority: High
Effort: Low

03 Data-Driven Market Analysis for Investors

Insight →
Clustering analysis revealed distinct micro-markets with varying pricing efficiency.
Recommendation →
Investors should adopt data-driven tools such as clustering and statistical analysis to identify undervalued areas instead of relying on intuition.
Expected Impact →
Higher ROI through identification of undervalued clusters and early investment opportunities.
Priority: High
Effort: Medium

SECTION THIRTEEN

13 Impact Estimation

RECOMMENDATION	IMPACT TYPE	EST. MAGNITUDE	LOGIC / ASSUMPTION
Invest in Top-10 Districts	Revenue	+40–60%	Based on current discount vs district average

Platform Tagging of Undervalued Zones	Efficiency	+18–25%	Based on buyer search intent signals
Market Timing Strategy	Cost Savings	−25–35%	Post-March dip (67% listing drop) creates buyer leverage → historical negotiation margin ≈ 12% average

SECTION FOURTEEN

14 Limitations

- Single-point snapshot — Data covers Jan–Mar 2026 only; seasonal fluctuations (especially North Coast, holiday properties) may distort annual patterns.
- Agent-submitted descriptions — Listing text may be exaggerated or templated, inflating sentiment scores for certain districts artificially.
- No transactional data — Listing price ≠ sale price; actual transaction values are unavailable, limiting real undervaluation confirmation.
- Sentiment model language — NLP applied to English/transliterated text only; Arabic-language descriptions processed with lower accuracy.
- Geospatial approximation — District-level lat/lon centroids used; intra-district variation not captured at street level.

SECTION FIFTEEN

15 Future Scope

MORE DATA

NEW DATA SOURCES

Historical listings (2020–2025) to enable price-change trend modelling and seasonality decomposition.

Actual transaction records (Aqarmap or government registries), Google Maps POI data (proximity to schools, metro) for amenity scoring.

PREDICTIVE MODEL

Time-series price forecasting (ARIMA / Prophet) per district; buyer-intent scoring model using NLP + behavioral signals.

REAL-TIME DASHBOARD

Automate weekly ETL pipeline via Airflow; connect live Tableau dashboard with PropertyFinder API refresh.

SECTION SIXTEEN

16 Conclusion

Cairo and Giza's residential property market lacks transparent, data-driven intelligence for identifying undervalued opportunities — a gap that leaves both investors and buyers dependent on agent perception rather than evidence. This project analysed 39,712 live PropertyFinder listings using a Python-based ETL and NLP pipeline, combining geospatial clustering with sentiment analysis to generate a composite value score per district. The key finding is that 462 micro-market zones — led by Coronado Beach, Ahmed Basha St. and Nadi Al Zamalek St. — show significant sentiment quality scores relative to their price-per-sqm, representing genuine undervaluation opportunities. The recommended action is for investors to prioritise these districts before broader market discovery closes the pricing gap, and for PropertyFinder to surface these zones as a product feature for value-seeking buyers.

A

APPENDIX

Appendix

- Data Dictionary — Full column definitions for all fields in the cleaned dataset
- Additional EDA Charts — Distribution plots, correlation heatmap, full property type breakdown not included in main report
- Python/SQL Logic Excerpts — Key cleaning and feature engineering code snippets
- Raw Cleaning Log — Output from notebooks/02_cleaning.ipynb showing before/after row counts per step

CM

MANDATORY

Contribution Matrix

MustmatchGitHub Insights,PRhistory,andcommitted files. Mismatches may resultinindividual grade adjustments.

TEAM MEMBER	DATASET & SOURCING	ETL & CLEANING	EDA & ANALYSIS	STATISTICAL ANALYSIS	TABLEAU DASHBOARD	REPORT WRITING	PPT & VIVA
Anshvardhan	✓	—	—	✓	✓	✓	—
Rahul	—	—	—	—	✓	—	✓
Prateek	—	—	—	✓	—	—	—
Ansh Sharma	—	—	✓	—	—	—	—

TEAM MEMBER	DATASET & SOURCING	ETL & CLEANING	EDA & ANALYSIS	STATISTICAL ANALYSIS	TABLEAU DASHBOARD	REPORT WRITING	PPT
Vivek	☑	☑	☑	☑	—	☑	☑
Prataya	—	—	—	—	—	—	—

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GitHub Repository · Tableau Public Dashboard